Database Management Systems

UE24CS351A

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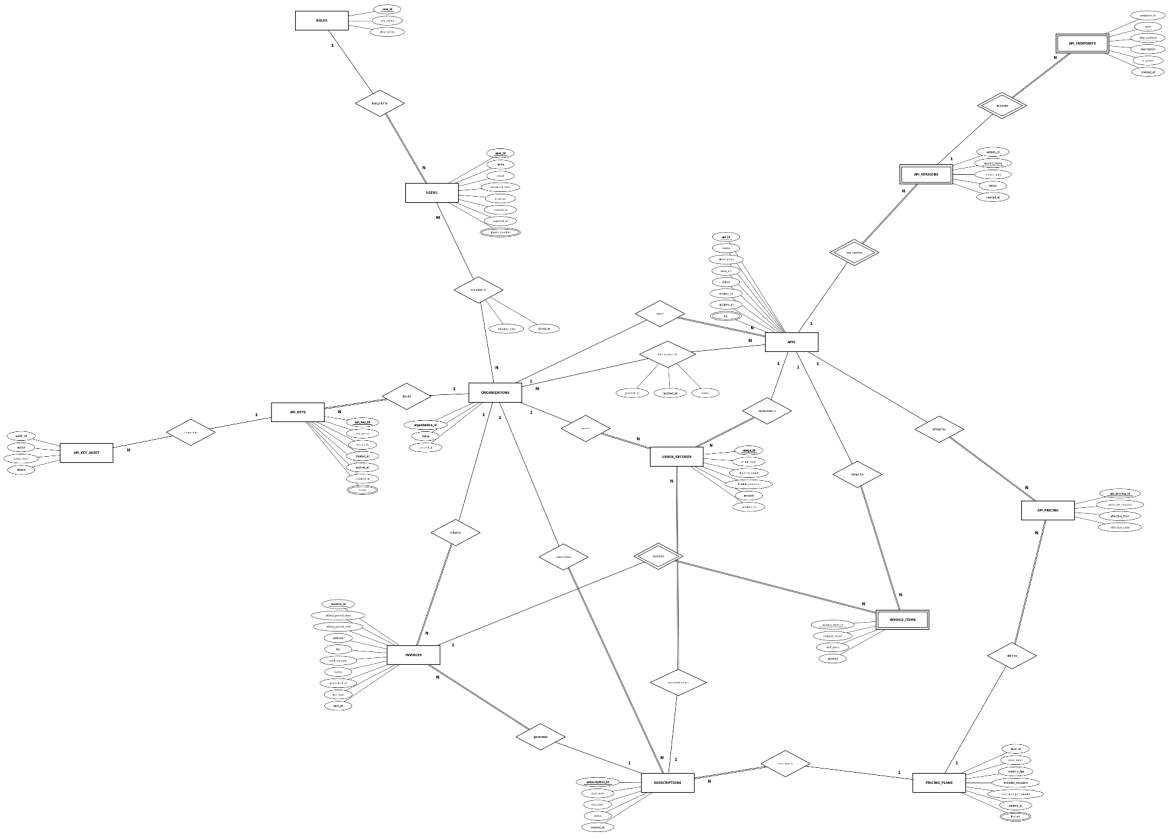
DBMS MINI PROJECT

PROBLEM STATEMENT:
Developer API usage and billing platform

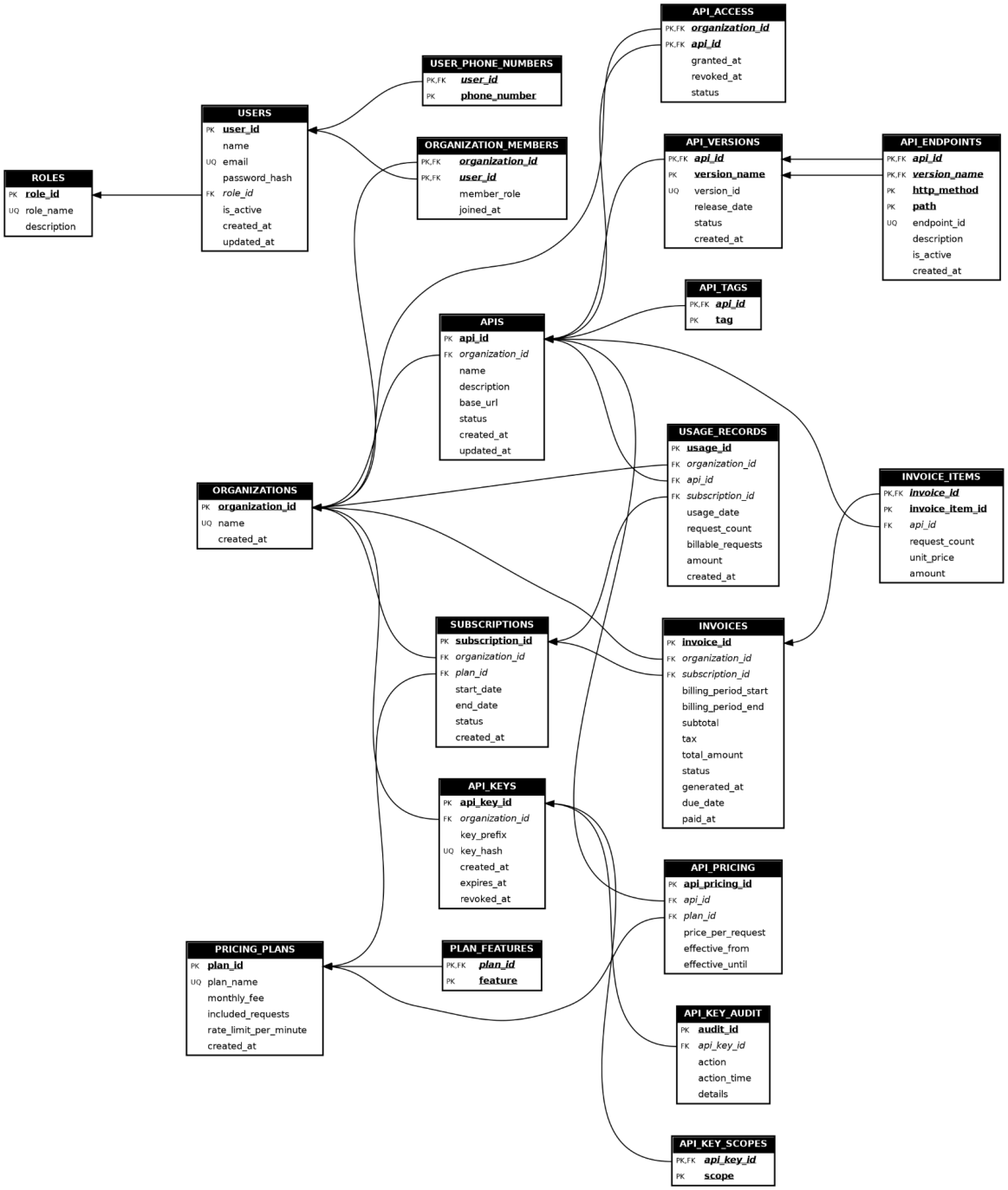
Organizations that publish APIs for other developers to use need a reliable way to control who can access those APIs, measure how much each customer uses them, and bill them correctly. Doing this by hand or with scattered spreadsheets leads to billing errors, unclear pricing and no record of who accessed what.

This project builds a relational database that manages users and organizations, API versions and endpoints, pricing plans and subscriptions, API keys and access control, daily usage records, and invoices with line items. Totals are calculated automatically, and key activity is logged for auditing.

ER DIAGRAM:


ROLES
assigned to
USERS
API_ENDPOINTS
access
API_VERSIONS
has version
APIS
has access to
ORGANIZATIONS
API_KEYS
belongs
API_KEY_AUDIT
USAGE_RECORDS
updated at
UPDATED
DEFINES
API_PRICING
INVOICES
INVOICE_ITEMS
subscribed
SUBSCRIPTIONS
PRICING_PLANS

RELATIONAL SCHEMA



ROLES(

role_id PK,

role_name UQ,

description

)

USERS(

user_id PK,

name,

email UQ,

password_hash,

role_id FK → ROLES.role_id,

is_active,

created_at,

updated_at

)

USER_PHONE_NUMBERS(-- multivalued attribute of USERS

user_id PK, FK → USERS.user_id,

phone_number PK

)

ORGANIZATIONS(

organization_id PK,

name UQ,

created_at

)

ORGANIZATION_MEMBERS(

organization_id PK, FK → ORGANIZATIONS.organization_id,

user_id PK, FK → USERS.user_id,

member_role,

joined_at

)

APIS(

api_id PK,

organization_id FK → ORGANIZATIONS.organization_id,

name,

description,

base_url,

status,

created_at,

updated_at

)

API_TAGS(-- multivalued attribute of APIS

api_id PK, FK → APIS.api_id,

tag PK

)

API_VERSIONS(-- weak entity, owner: APIS

api_id PK, FK → APIS.api_id,

version_name PK, -- partial key

version_id UQ,

release_date,

status,

created_at

)

API_ENDPOINTS(-- weak entity, owner: API_VERSIONS

api_id PK,

version_name PK,

http_method PK, -- partial key

path PK, -- partial key

endpoint_id UQ,

description,

is_active,

created_at,

FK (api_id, version_name) → API_VERSIONS(api_id, version_name)

)

PRICING_PLANS(

plan_id PK,

plan_name UQ,

monthly_fee,

included_requests,

rate_limit_per_minute,

created_at

)

PLAN_FEATURES(-- multivalued attribute of PRICING_PLANS

plan_id PK, FK → PRICING_PLANS.plan_id,

feature PK

)

API_PRICING(

api_pricing_id PK,

api_id FK → APIS.api_id,

plan_id FK → PRICING_PLANS.plan_id,

price_per_request,

effective_from,

effective_until

)

SUBSCRIPTIONS(

subscription_id PK,

organization_id FK → ORGANIZATIONS.organization_id,

plan_id FK → PRICING_PLANS.plan_id,

start_date,

end_date,

status,

created_at

)

API_KEYS(

api_key_id PK,

organization_id FK → ORGANIZATIONS.organization_id,

key_prefix,

key_hash UQ,

created_at,

expires_at,

revoked_at

)

API_KEY_SCOPES(-- multivalued attribute of API_KEYS

api_key_id PK, FK → API_KEYS.api_key_id,

scope PK

)

API_ACCESS(

organization_id PK, FK → ORGANIZATIONS.organization_id,

api_id PK, FK → APIS.api_id,

granted_at,

revoked_at,

status

)

USAGE_RECORDS(

usage_id PK,

organization_id FK → ORGANIZATIONS.organization_id,

api_id FK → APIS.api_id,

subscription_id FK → SUBSCRIPTIONS.subscription_id,

usage_date,

request_count,

billable_requests,

amount,

created_at,

UQ (organization_id, api_id, subscription_id, usage_date)

)

INVOICES(

invoice_id PK,

organization_id FK → ORGANIZATIONS.organization_id,

subscription_id FK → SUBSCRIPTIONS.subscription_id,

billing_period_start,

billing_period_end,

subtotal,

tax,

total_amount,

status,

generated_at,

due_date,

paid_at

)

INVOICE_ITEMS(-- weak entity, owner: INVOICES

invoice_id PK, FK → INVOICES.invoice_id,

invoice_item_id PK, -- partial key (line number)

api_id FK → APIS.api_id,

request_count,

unit_price,

amount

)

API_KEY_AUDIT(

audit_id PK,

api_key_id FK → API_KEYS.api_key_id (nullable),

action,

action_time,

details

)

DDL COMMANDS

```
mysql> /*
   /*> ========================================================================
   /*> 2. ROLES                                          (strong entity)
   /*> ========================================================================
   /*> Relationship: roles 1 ---- N users   ("assigned to")
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE roles (
    ->     role_id     BIGINT      NOT NULL AUTO_INCREMENT,
    ->     role_name   VARCHAR(50) NOT NULL,
    ->     description TEXT,
    ->
    ->     PRIMARY KEY (role_id),
    ->     UNIQUE KEY uq_roles_name (role_name)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql>
mysql> /*
   /*> ========================================================================
   /*> 3. USERS                                          (strong entity)
   /*> ========================================================================
   /*> "assigned to": every user has exactly one role  -> role_id NOT NULL
   /*> (total participation of USERS)
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE users (
    ->     user_id       BIGINT       NOT NULL AUTO_INCREMENT,
    ->     name          VARCHAR(100) NOT NULL,
    ->     email         VARCHAR(255) NOT NULL,
    ->     password_hash TEXT         NOT NULL,
    ->     role_id       BIGINT       NOT NULL,
    ->     is_active     BOOLEAN      NOT NULL DEFAULT TRUE,
    ->     created_at    DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->     updated_at    DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP
    ->                                ON UPDATE CURRENT_TIMESTAMP,
    ->
    ->     PRIMARY KEY (user_id),
    ->     UNIQUE KEY uq_users_email (email),
    ->
    ->     CONSTRAINT fk_users_role
    ->         FOREIGN KEY (role_id) REFERENCES roles (role_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> /*
   /*> ========================================================================
   /*> 4. USER PHONE NUMBERS            (multivalued attribute of USERS)
   /*> ========================================================================
   /*> A user can have many phone numbers -> separate table.
   /*> PK = (user_id, phone_number): the owner's PK + the attribute value.
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE user_phone_numbers (
    ->     user_id      BIGINT      NOT NULL,
    ->     phone_number VARCHAR(20) NOT NULL,
    ->
    ->     PRIMARY KEY (user_id, phone_number),
    ->
    ->     CONSTRAINT fk_user_phone_user
    ->         FOREIGN KEY (user_id) REFERENCES users (user_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql>
mysql> /*
   /*> ========================================================================
   /*> 5. ORGANIZATIONS                                  (strong entity)
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE organizations (
    ->     organization_id BIGINT       NOT NULL AUTO_INCREMENT,
    ->     name            VARCHAR(150) NOT NULL,
    ->     created_at      DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->
    ->     PRIMARY KEY (organization_id),
    ->     UNIQUE KEY uq_organizations_name (name)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.02 sec)
```

```
   /*> ========================================================================
   /*> 6. ORGANIZATION MEMBERS       (M:N relationship "member of" with attributes)
   /*> ========================================================================
   /*> users M ---- N organizations.  Both sides are partial participation.
   /*> Relationship attributes: member_role, joined_at.
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE organization_members (
    ->     organization_id BIGINT      NOT NULL,
    ->     user_id         BIGINT      NOT NULL,
    ->     member_role     VARCHAR(50) NOT NULL DEFAULT 'DEVELOPER',
    ->     joined_at       DATETIME    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->
    ->     PRIMARY KEY (organization_id, user_id),
    ->
    ->     CONSTRAINT fk_org_members_org
    ->         FOREIGN KEY (organization_id) REFERENCES organizations (organization_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT fk_org_members_user
    ->         FOREIGN KEY (user_id) REFERENCES users (user_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT chk_member_role
    ->         CHECK (member_role IN ('OWNER', 'ADMIN', 'DEVELOPER', 'VIEWER'))
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql>
mysql> /*
   /*> ========================================================================
   /*> 7. APIS                                          (strong entity)
   /*> ========================================================================
   /*> "owns": organizations 1 ---- N apis.  Every API has exactly one organization
   /*> (total participation of APIS) -> organization_id NOT NULL.
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE apis (
    ->     api_id          BIGINT       NOT NULL AUTO_INCREMENT,
    ->     organization_id BIGINT       NOT NULL,
    ->     name            VARCHAR(150) NOT NULL,
    ->     description     TEXT,
    ->     base_url        TEXT         NOT NULL,
    ->     status          VARCHAR(20)  NOT NULL DEFAULT 'ACTIVE',
    ->     created_at      DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->     updated_at      DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP
    ->                                  ON UPDATE CURRENT_TIMESTAMP,
    ->
    ->     PRIMARY KEY (api_id),
    ->
    ->     CONSTRAINT fk_apis_organization
    ->         FOREIGN KEY (organization_id) REFERENCES organizations (organization_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT chk_api_status
    ->         CHECK (status IN ('ACTIVE', 'INACTIVE', 'DEPRECATED'))
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql>
mysql> /*
   /*> ========================================================================
   /*> 8. API TAGS                          (multivalued attribute of APIS)
   /*> ========================================================================
   /*> An API can carry many tags (e.g. 'nlp', 'translation', 'beta').
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE api_tags (
    ->     api_id BIGINT      NOT NULL,
    ->     tag    VARCHAR(50) NOT NULL,
    ->
    ->     PRIMARY KEY (api_id, tag),
    ->
    ->     CONSTRAINT fk_api_tags_api
    ->         FOREIGN KEY (api_id) REFERENCES apis (api_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql>
mysql> /*
   /*> ========================================================================
   /*> 9. API VERSIONS            (WEAK ENTITY, owner = APIS, "has version")
   /*> ========================================================================
   /*> Identifying relationship: apis 1 ---- N api_versions.
   /*>
   /*>     Primary key  = owner PK  +  partial key
   /*>                  = api_id    +  version_name
   /*>
   /*> A version name such as 'v1' only has meaning inside one API, so the version
   /*> is identified by the API's key together with its own (partial) key.
   /*>
   /*> version_id is kept as an ordinary attribute of the diagram; it is a surrogate
   /*> value (UNIQUE), not the primary key.
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE api_versions (
    ->     api_id       BIGINT      NOT NULL,          -- owner PK (part of PK + FK)
    ->     version_name VARCHAR(30) NOT NULL,          -- partial key (part of PK)
    ->     version_id   BIGINT      NOT NULL AUTO_INCREMENT,
    ->     release_date DATE        NOT NULL DEFAULT (CURRENT_DATE),
    ->     status       VARCHAR(20) NOT NULL DEFAULT 'ACTIVE',
    ->     created_at   DATETIME    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->
    ->     PRIMARY KEY (api_id, version_name),
    ->     UNIQUE KEY uq_api_versions_version_id (version_id),
    ->
    ->     CONSTRAINT fk_versions_api
    ->         FOREIGN KEY (api_id) REFERENCES apis (api_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT chk_version_status
    ->         CHECK (status IN ('ACTIVE', 'INACTIVE', 'DEPRECATED'))
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> /*
   /*> ========================================================================
   /*> 10. API ENDPOINTS      (WEAK ENTITY, owner = API_VERSIONS, "exposes")
   /*> ========================================================================
   /*> Identifying relationship: api_versions 1 ---- N api_endpoints.
   /*>
   /*>     Primary key  = owner PK              +  partial key
   /*>                  = (api_id, version_name) +  (http_method, path)
   /*>
   /*> The whole composite key of the owner (api_id, version_name) is copied into the
   /*> child, and the same pair is the composite foreign key. Two endpoints of the
   /*> same API version can never share the same method + path.
   /*>
   /*> endpoint_id is kept as an ordinary attribute of the diagram (surrogate,
   /*> UNIQUE), not the primary key.
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE api_endpoints (
    ->     api_id       BIGINT       NOT NULL,         -- owner PK (part of PK + FK)
    ->     version_name VARCHAR(30)  NOT NULL,         -- owner PK (part of PK + FK)
    ->     http_method  VARCHAR(10)  NOT NULL,         -- partial key
    ->     path         VARCHAR(255) NOT NULL,         -- partial key
    ->     endpoint_id  BIGINT       NOT NULL AUTO_INCREMENT,
    ->     description  TEXT,
    ->     is_active    BOOLEAN      NOT NULL DEFAULT TRUE,
    ->     created_at   DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->
    ->     PRIMARY KEY (api_id, version_name, http_method, path),
    ->     UNIQUE KEY uq_api_endpoints_endpoint_id (endpoint_id),
    ->
    ->     CONSTRAINT fk_endpoint_version
    ->         FOREIGN KEY (api_id, version_name)
    ->         REFERENCES api_versions (api_id, version_name)
    ->         ON DELETE CASCADE ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT chk_http_method
    ->         CHECK (http_method IN
    ->                ('GET', 'POST', 'PUT', 'PATCH', 'DELETE', 'HEAD', 'OPTIONS'))
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> /*
   /*> ========================================================================
   /*> 11. PRICING PLANS                                    (strong entity)
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE pricing_plans (
    ->     plan_id              BIGINT        NOT NULL AUTO_INCREMENT,
    ->     plan_name            VARCHAR(100)  NOT NULL,
    ->     monthly_fee          DECIMAL(12,2) NOT NULL DEFAULT 0,
    ->     included_requests    BIGINT        NOT NULL DEFAULT 0,
    ->     rate_limit_per_minute INT          NOT NULL DEFAULT 60,
    ->     created_at           DATETIME      NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->
    ->     PRIMARY KEY (plan_id),
    ->     UNIQUE KEY uq_pricing_plans_name (plan_name),
    ->
    ->     CONSTRAINT chk_monthly_fee       CHECK (monthly_fee >= 0),
    ->     CONSTRAINT chk_included_requests CHECK (included_requests >= 0),
    ->     CONSTRAINT chk_rate_limit        CHECK (rate_limit_per_minute > 0)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql>
mysql> /*
   /*> ========================================================================
   /*> 12. PLAN FEATURES               (multivalued attribute of PRICING_PLANS)
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE plan_features (
    ->     plan_id BIGINT       NOT NULL,
    ->     feature VARCHAR(100) NOT NULL,
    ->
    ->     PRIMARY KEY (plan_id, feature),
    ->
    ->     CONSTRAINT fk_plan_features_plan
    ->         FOREIGN KEY (plan_id) REFERENCES pricing_plans (plan_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE api_pricing (
    ->     api_pricing_id    BIGINT        NOT NULL AUTO_INCREMENT,
    ->     api_id            BIGINT        NOT NULL,
    ->     plan_id           BIGINT        NOT NULL,
    ->     price_per_request DECIMAL(12,4) NOT NULL,
    ->     effective_from    DATETIME      NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->     effective_until   DATETIME      NULL,
    ->
    ->     PRIMARY KEY (api_pricing_id),
    ->
    ->     CONSTRAINT fk_api_pricing_api
    ->         FOREIGN KEY (api_id) REFERENCES apis (api_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT fk_api_pricing_plan
    ->         FOREIGN KEY (plan_id) REFERENCES pricing_plans (plan_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT chk_price_per_request CHECK (price_per_request >= 0),
    ->
    ->     CONSTRAINT chk_pricing_dates
    ->         CHECK (effective_until IS NULL OR effective_until > effective_from)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql>
mysql> /*
   /*> ========================================================================
   /*> 14. SUBSCRIPTIONS                                    (strong entity)
   /*> ========================================================================
   /*> "subscribes":  organizations  1 ---- N subscriptions  (total for subscriptions)
   /*> "selected in": pricing_plans  1 ---- N subscriptions  (total for subscriptions)
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE subscriptions (
    ->     subscription_id BIGINT      NOT NULL AUTO_INCREMENT,
    ->     organization_id BIGINT      NOT NULL,
    ->     plan_id         BIGINT      NOT NULL,
    ->     start_date      DATE        NOT NULL,
    ->     end_date        DATE        NULL,
    ->     status          VARCHAR(20) NOT NULL DEFAULT 'ACTIVE',
    ->     created_at      DATETIME    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->
    ->     PRIMARY KEY (subscription_id),
    ->
    ->     CONSTRAINT fk_subscription_org
    ->         FOREIGN KEY (organization_id) REFERENCES organizations (organization_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT fk_subscription_plan
    ->         FOREIGN KEY (plan_id) REFERENCES pricing_plans (plan_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT chk_subscription_status
    ->         CHECK (status IN ('ACTIVE', 'PAUSED', 'CANCELLED', 'EXPIRED')),
    ->
    ->     CONSTRAINT chk_subscription_dates
    ->         CHECK (end_date IS NULL OR end_date >= start_date)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> /*
   /*> ========================================================================
   /*> 15. API KEYS                                          (strong entity)
   /*> ========================================================================
   /*> "issues": organizations 1 ---- N api_keys (total for api_keys).
   /*> Only the hash of a key is stored, never the key itself.
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE api_keys (
    ->     api_key_id      BIGINT       NOT NULL AUTO_INCREMENT,
    ->     organization_id BIGINT       NOT NULL,
    ->     key_prefix      VARCHAR(20)  NOT NULL,
    ->     key_hash        VARCHAR(255) NOT NULL,
    ->     created_at      DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->     expires_at      DATETIME     NULL,
    ->     revoked_at      DATETIME     NULL,
    ->
    ->     PRIMARY KEY (api_key_id),
    ->     UNIQUE KEY uq_api_keys_hash (key_hash),
    ->
    ->     CONSTRAINT fk_api_key_org
    ->         FOREIGN KEY (organization_id) REFERENCES organizations (organization_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT chk_key_expiry
    ->         CHECK (expires_at IS NULL OR expires_at > created_at)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql>
mysql> /*
   /*> ========================================================================
   /*> 16. API KEY SCOPES                 (multivalued attribute of API_KEYS)
   /*> ========================================================================
   /*> A key can be limited to several scopes (e.g. 'read', 'write', 'billing').
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE api_key_scopes (
    ->     api_key_id BIGINT      NOT NULL,
    ->     scope      VARCHAR(50) NOT NULL,
    ->
    ->     PRIMARY KEY (api_key_id, scope),
    ->
    ->     CONSTRAINT fk_key_scopes_key
    ->         FOREIGN KEY (api_key_id) REFERENCES api_keys (api_key_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> /*
   /*> ========================================================================
   /*> 17. API ACCESS            (M:N relationship "has access to" with attributes)
   /*> ========================================================================
   /*> organizations M ---- N apis.  Both sides are partial participation.
   /*> Relationship attributes: granted_at, revoked_at, status.
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE api_access (
    ->     organization_id BIGINT      NOT NULL,
    ->     api_id          BIGINT      NOT NULL,
    ->     granted_at      DATETIME    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->     revoked_at      DATETIME    NULL,
    ->     status          VARCHAR(20) NOT NULL DEFAULT 'ACTIVE',
    ->
    ->     PRIMARY KEY (organization_id, api_id),
    ->
    ->     CONSTRAINT fk_api_access_org
    ->         FOREIGN KEY (organization_id) REFERENCES organizations (organization_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT fk_api_access_api
    ->         FOREIGN KEY (api_id) REFERENCES apis (api_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT chk_access_status
    ->         CHECK (status IN ('ACTIVE', 'REVOKED')),
    ->
    ->     CONSTRAINT chk_access_dates
    ->         CHECK (revoked_at IS NULL OR revoked_at >= granted_at)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> /*
   /*> ========================================================================
   /*> 18. USAGE RECORDS                                    (strong entity)
   /*> ========================================================================
   /*> Three total-participation relationships:
   /*>     "incurs"        organizations 1 ---- N usage_records
   /*>     "consumed in"   apis          1 ---- N usage_records
   /*>     "metered under" subscriptions 1 ---- N usage_records
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE usage_records (
    ->     usage_id          BIGINT        NOT NULL AUTO_INCREMENT,
    ->     organization_id   BIGINT        NOT NULL,
    ->     api_id            BIGINT        NOT NULL,
    ->     subscription_id   BIGINT        NOT NULL,
    ->     usage_date        DATE          NOT NULL,
    ->     request_count     BIGINT        NOT NULL DEFAULT 0,
    ->     billable_requests BIGINT        NOT NULL DEFAULT 0,
    ->     amount            DECIMAL(12,2) NOT NULL DEFAULT 0,
    ->     created_at        DATETIME      NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->
    ->     PRIMARY KEY (usage_id),
    ->
    ->     CONSTRAINT uq_daily_usage
    ->         UNIQUE (organization_id, api_id, subscription_id, usage_date),
    ->
    ->     CONSTRAINT fk_usage_org
    ->         FOREIGN KEY (organization_id) REFERENCES organizations (organization_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT fk_usage_api
    ->         FOREIGN KEY (api_id) REFERENCES apis (api_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT fk_usage_subscription
    ->         FOREIGN KEY (subscription_id) REFERENCES subscriptions (subscription_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT chk_request_count CHECK (request_count >= 0),
    ->
    ->     CONSTRAINT chk_billable_requests
    ->         CHECK (billable_requests >= 0 AND billable_requests <= request_count),
    ->
    ->     CONSTRAINT chk_usage_amount CHECK (amount >= 0)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> /*
   /*> ========================================================================
   /*> 19. INVOICES                                        (strong entity)
   /*> ========================================================================
   /*> "billed to":  organizations 1 ---- N invoices  (total for invoices)
   /*> "generates":  subscriptions 1 ---- N invoices  (total for invoices)
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE invoices (
    ->     invoice_id          BIGINT        NOT NULL AUTO_INCREMENT,
    ->     organization_id     BIGINT        NOT NULL,
    ->     subscription_id     BIGINT        NOT NULL,
    ->     billing_period_start DATE         NOT NULL,
    ->     billing_period_end  DATE          NOT NULL,
    ->     subtotal            DECIMAL(12,2) NOT NULL DEFAULT 0,
    ->     tax                 DECIMAL(12,2) NOT NULL DEFAULT 0,
    ->     total_amount        DECIMAL(12,2) NOT NULL DEFAULT 0,
    ->     status              VARCHAR(20)   NOT NULL DEFAULT 'DRAFT',
    ->     generated_at        DATETIME      NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->     due_date            DATE          NULL,
    ->     paid_at             DATETIME      NULL,
    ->
    ->     PRIMARY KEY (invoice_id),
    ->
    ->     CONSTRAINT fk_invoice_org
    ->         FOREIGN KEY (organization_id) REFERENCES organizations (organization_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT fk_invoice_subscription
    ->         FOREIGN KEY (subscription_id) REFERENCES subscriptions (subscription_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT chk_invoice_period   CHECK (billing_period_end >= billing_period_start),
    ->     CONSTRAINT chk_invoice_subtotal CHECK (subtotal >= 0),
    ->     CONSTRAINT chk_invoice_tax      CHECK (tax >= 0),
    ->     CONSTRAINT chk_invoice_total    CHECK (total_amount >= 0),
    ->
    ->     CONSTRAINT chk_invoice_status
    ->         CHECK (status IN ('DRAFT', 'ISSUED', 'PAID', 'OVERDUE', 'CANCELLED'))
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.05 sec)
```

```
*/
mysql>
mysql> CREATE TABLE invoice_items (
    ->     invoice_id      BIGINT        NOT NULL,             -- owner PK (part of PK + FK)
    ->     invoice_item_id INT           NOT NULL DEFAULT 0,   -- partial key (line no.)
    ->     api_id          BIGINT        NOT NULL,
    ->     request_count   BIGINT        NOT NULL,
    ->     unit_price      DECIMAL(12,4) NOT NULL,
    ->     amount          DECIMAL(12,2) NOT NULL,
    ->
    ->     PRIMARY KEY (invoice_id, invoice_item_id),
    ->
    ->     CONSTRAINT fk_invoice_item_invoice
    ->         FOREIGN KEY (invoice_id) REFERENCES invoices (invoice_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT fk_invoice_item_api
    ->         FOREIGN KEY (api_id) REFERENCES apis (api_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->
    ->     CONSTRAINT chk_invoice_item_requests CHECK (request_count >= 0),
    ->     CONSTRAINT chk_invoice_item_price    CHECK (unit_price >= 0),
    ->     CONSTRAINT chk_invoice_item_amount   CHECK (amount >= 0)
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql>
mysql> /*
   /*> ========================================================================
   /*> 21. API KEY AUDIT                                    (strong entity)
   /*> ========================================================================
   /*> "logged in": api_keys 1 ---- N api_key_audit.
   /*> PARTIAL participation of the audit row: api_key_id is nullable and the
   /*> foreign key is ON DELETE SET NULL, so the audit history survives even when a
   /*> key row is deleted.
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE TABLE api_key_audit (
    ->     audit_id    BIGINT      NOT NULL AUTO_INCREMENT,
    ->     api_key_id  BIGINT      NULL,
    ->     action      VARCHAR(30) NOT NULL,
    ->     action_time DATETIME    NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->     details     TEXT,
    ->
    ->     PRIMARY KEY (audit_id),
    ->
    ->     CONSTRAINT fk_audit_api_key
    ->         FOREIGN KEY (api_key_id) REFERENCES api_keys (api_key_id)
    ->         ON DELETE SET NULL ON UPDATE CASCADE
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.03 sec)
```

```
   /*> ========================================================================
   /*> 22. INDEXES
   /*> ========================================================================
   /*> Primary keys, UNIQUE constraints and foreign keys already create indexes in
   /*> MySQL. The indexes below are additional ones for common lookups.
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE INDEX idx_users_active         ON users (is_active);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_org_members_user     ON organization_members (user_id);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_apis_status          ON apis (status);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_endpoints_method_path ON api_endpoints (http_method, path);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_api_pricing_api_plan  ON api_pricing (api_id, plan_id);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_subscriptions_status  ON subscriptions (status);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_api_keys_prefix       ON api_keys (key_prefix);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_api_access_status     ON api_access (status);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_usage_org_date        ON usage_records (organization_id, usage_date);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_usage_api_date        ON usage_records (api_id, usage_date);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_usage_date            ON usage_records (usage_date);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_invoices_status       ON invoices (status);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_invoices_period       ON invoices (billing_period_start, billing_period_end);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_invoice_items_api     ON invoice_items (api_id);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_api_tags_tag          ON api_tags (tag);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_key_scopes_scope      ON api_key_scopes (scope);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
   /*> ========================================================================
   /*> 23. TRIGGERS AND PROCEDURE
   /*> ========================================================================
   /*>     - users.updated_at / apis.updated_at use ON UPDATE CURRENT_TIMESTAMP in
   /*>       the table definitions (no trigger needed in MySQL).
   /*>     - trg_invoice_item_number : numbers the lines of each invoice (weak-entity
   /*>                                 partial key).
   /*>     - refresh_invoice_total   : recomputes subtotal and total of an invoice.
   /*>     - trg_invoice_items_*     : call the procedure after INSERT/UPDATE/DELETE.
   /*>     - trg_api_key_*           : write CREATED / REVOKED rows to api_key_audit.
   /*> ========================================================================
   /*> */
mysql>
mysql> DELIMITER $$
mysql>
mysql> -- 23.1 Partial key generator for the weak entity invoice_items
mysql> CREATE TRIGGER trg_invoice_item_number
    -> BEFORE INSERT ON invoice_items
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.invoice_item_id IS NULL OR NEW.invoice_item_id = 0 THEN
    ->         SET NEW.invoice_item_id =
    ->             (SELECT COALESCE(MAX(invoice_item_id), 0) + 1
    ->              FROM invoice_items
    ->              WHERE invoice_id = NEW.invoice_id);
    ->     END IF;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql>
mysql> -- 23.2 Invoice total recalculation
mysql> CREATE PROCEDURE refresh_invoice_total(IN p_invoice_id BIGINT)
    -> BEGIN
    ->     UPDATE invoices
    ->     SET subtotal     = (SELECT COALESCE(SUM(amount), 0)
    ->                         FROM invoice_items
    ->                         WHERE invoice_id = p_invoice_id),
    ->         total_amount = (SELECT COALESCE(SUM(amount), 0)
    ->                         FROM invoice_items
    ->                         WHERE invoice_id = p_invoice_id) + tax
    ->     WHERE invoice_id = p_invoice_id;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CREATE TRIGGER trg_invoice_items_ai
    -> AFTER INSERT ON invoice_items
    -> FOR EACH ROW
    -> BEGIN
    ->     CALL refresh_invoice_total(NEW.invoice_id);
    -> END$$
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TRIGGER trg_invoice_items_au
    -> AFTER UPDATE ON invoice_items
    -> FOR EACH ROW
    -> BEGIN
    ->     CALL refresh_invoice_total(NEW.invoice_id);
    ->     IF OLD.invoice_id <> NEW.invoice_id THEN
    ->         CALL refresh_invoice_total(OLD.invoice_id);
    ->     END IF;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CREATE TRIGGER trg_invoice_items_ad
    -> AFTER DELETE ON invoice_items
    -> FOR EACH ROW
    -> BEGIN
    ->     CALL refresh_invoice_total(OLD.invoice_id);
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql>
mysql> -- 23.3 API key audit
mysql> CREATE TRIGGER trg_api_key_ai
    -> AFTER INSERT ON api_keys
    -> FOR EACH ROW
    -> BEGIN
    ->     INSERT INTO api_key_audit (api_key_id, action, details)
    ->     VALUES (NEW.api_key_id, 'CREATED', 'API key created');
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CREATE TRIGGER trg_api_key_au
    -> AFTER UPDATE ON api_keys
    -> FOR EACH ROW
    -> BEGIN
    ->     IF OLD.revoked_at IS NULL AND NEW.revoked_at IS NOT NULL THEN
    ->         INSERT INTO api_key_audit (api_key_id, action, details)
    ->         VALUES (NEW.api_key_id, 'REVOKED', 'API key revoked');
    ->     END IF;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql>
```

```
   /*> ========================================================================
   /*> 25. VIEWS
   /*> ========================================================================
   /*> */
mysql>
mysql> CREATE OR REPLACE VIEW api_usage_summary AS
    -> SELECT
    ->     a.api_id,
    ->     a.name                  AS api_name,
    ->     SUM(u.request_count)    AS total_requests,
    ->     SUM(u.billable_requests) AS total_billable_requests,
    ->     SUM(u.amount)           AS total_amount,
    ->     AVG(u.amount)           AS average_usage_amount
    -> FROM usage_records u
    -> JOIN apis a ON u.api_id = a.api_id
    -> GROUP BY a.api_id, a.name;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql>
mysql> CREATE OR REPLACE VIEW monthly_api_usage AS
    -> SELECT
    ->     u.organization_id,
    ->     o.name                   AS organization_name,
    ->     u.api_id,
    ->     a.name                   AS api_name,
    ->     DATE_FORMAT(u.usage_date, '%Y-%m-01') AS month,
    ->     SUM(u.request_count)     AS total_requests,
    ->     SUM(u.billable_requests) AS billable_requests,
    ->     SUM(u.amount)            AS total_amount
    -> FROM usage_records u
    -> JOIN organizations o ON u.organization_id = o.organization_id
    -> JOIN apis a          ON u.api_id = a.api_id
    -> GROUP BY u.organization_id, o.name, u.api_id, a.name,
    ->          DATE_FORMAT(u.usage_date, '%Y-%m-01');
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql>
mysql> CREATE OR REPLACE VIEW organization_billing_summary AS
    -> SELECT
    ->     o.organization_id,
    ->     o.name                                AS organization_name,
    ->     COUNT(DISTINCT i.invoice_id)          AS invoice_count,
    ->     COALESCE(SUM(i.subtotal), 0)          AS subtotal,
    ->     COALESCE(SUM(i.tax), 0)               AS tax,
    ->     COALESCE(SUM(i.total_amount), 0)      AS total_billed
    -> FROM organizations o
    -> LEFT JOIN invoices i ON o.organization_id = i.organization_id
    -> GROUP BY o.organization_id, o.name;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> /*
   /*> ========================================================================
   /*> 26. STORED FUNCTIONS
   /*> ========================================================================
   /*> calculate_usage_cost:
   /*>     active subscription -> plan allowance -> current API price ->
   /*>     (requests - included) x price.
   /*> calculate_billable_requests:
   /*>     requests beyond the plan allowance (never negative).
   /*> ========================================================================
   /*> */
mysql>
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION calculate_usage_cost(
    ->     p_organization_id BIGINT,
    ->     p_api_id          BIGINT,
    ->     p_request_count   BIGINT
    -> )
    -> RETURNS DECIMAL(12,2)
    -> READS SQL DATA
    -> BEGIN
    ->     DECLARE v_plan_id           BIGINT        DEFAULT NULL;
    ->     DECLARE v_included_requests BIGINT        DEFAULT NULL;
    ->     DECLARE v_price             DECIMAL(12,4) DEFAULT NULL;
    ->     DECLARE v_billable_requests BIGINT;
    ->
    ->     SELECT plan_id INTO v_plan_id
    ->     FROM subscriptions
    ->     WHERE organization_id = p_organization_id
    ->       AND status = 'ACTIVE'
    ->     ORDER BY start_date DESC
    ->     LIMIT 1;
    ->
    ->     IF v_plan_id IS NULL THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'No active subscription found';
    ->     END IF;
    ->
    ->     SELECT included_requests INTO v_included_requests
    ->     FROM pricing_plans
    ->     WHERE plan_id = v_plan_id;
    ->
    ->     SELECT price_per_request INTO v_price
    ->     FROM api_pricing
    ->     WHERE api_id = p_api_id
    ->       AND plan_id = v_plan_id
    ->       AND effective_from <= CURRENT_TIMESTAMP
    ->       AND (effective_until IS NULL OR effective_until > CURRENT_TIMESTAMP)
    ->     ORDER BY effective_from DESC
    ->     LIMIT 1;
    ->
    ->     IF v_price IS NULL THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'No pricing found for this API and plan';
    ->     END IF;
    ->
    ->     SET v_billable_requests = GREATEST(0, p_request_count - v_included_requests);
    ->
    ->     RETURN ROUND(v_billable_requests * v_price, 2);
    -> END$$
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE FUNCTION calculate_billable_requests(
    ->     p_organization_id BIGINT,
    ->     p_request_count   BIGINT
    -> )
    -> RETURNS BIGINT
    -> READS SQL DATA
    -> BEGIN
    ->     DECLARE v_included_requests BIGINT DEFAULT NULL;
    ->
    ->     SELECT pp.included_requests INTO v_included_requests
    ->     FROM subscriptions s
    ->     JOIN pricing_plans pp ON s.plan_id = pp.plan_id
    ->     WHERE s.organization_id = p_organization_id
    ->       AND s.status = 'ACTIVE'
    ->     ORDER BY s.start_date DESC
    ->     LIMIT 1;
    ->
    ->     IF v_included_requests IS NULL THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'No active subscription found';
    ->     END IF;
    ->
    ->     RETURN GREATEST(0, p_request_count - v_included_requests);
    -> END$$
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> DELIMITER ;
mysql>
```

DML COMMANDS

```
mysql> /*
   /*> ========================================================================
   /*> 24. SAMPLE DATA
   /*> ========================================================================
   /*> */
mysql>
mysql> INSERT INTO roles (role_name, description) VALUES
    ->     ('ADMIN',     'Platform administrator'),
    ->     ('DEVELOPER', 'API consumer/developer');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO users (name, email, password_hash, role_id) VALUES
    ->     ('System Administrator', 'admin@example.com',
    ->      '$2b$12$example_admin_hash',
    ->      (SELECT role_id FROM roles WHERE role_name = 'ADMIN')),
    ->     ('John Developer', 'john@example.com',
    ->      '$2b$12$example_developer_hash',
    ->      (SELECT role_id FROM roles WHERE role_name = 'DEVELOPER'));
Query OK, 2 rows affected (0.00 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql>
mysql> -- multivalued attribute: phone numbers
mysql> INSERT INTO user_phone_numbers (user_id, phone_number) VALUES
    ->     ((SELECT user_id FROM users WHERE email = 'admin@example.com'), '+91-9000000001'),
    ->     ((SELECT user_id FROM users WHERE email = 'admin@example.com'), '+91-9000000002'),
    ->     ((SELECT user_id FROM users WHERE email = 'john@example.com'),  '+91-9000000003');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO organizations (name) VALUES
    ->     ('Acme Technologies'),
    ->     ('DevCloud Solutions');
Query OK, 2 rows affected (0.00 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO organization_members (organization_id, user_id, member_role) VALUES
    ->     ((SELECT organization_id FROM organizations WHERE name = 'Acme Technologies'),
    ->      (SELECT user_id FROM users WHERE email = 'admin@example.com'), 'OWNER'),
    ->     ((SELECT organization_id FROM organizations WHERE name = 'Acme Technologies'),
    ->      (SELECT user_id FROM users WHERE email = 'john@example.com'),  'DEVELOPER');
Query OK, 2 rows affected (0.00 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO apis (organization_id, name, description, base_url) VALUES
    ->     ((SELECT organization_id FROM organizations WHERE name = 'Acme Technologies'),
    ->      'Translation API', 'Text translation service', 'https://api.example.com/translate'),
    ->     ((SELECT organization_id FROM organizations WHERE name = 'Acme Technologies'),
    ->      'Image Processing API', 'Image processing service', 'https://api.example.com/image'),
    ->     ((SELECT organization_id FROM organizations WHERE name = 'Acme Technologies'),
    ->      'Report Generation API', 'Generates PDF reports', 'https://api.example.com/reports');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> -- multivalued attribute: tags
mysql> INSERT INTO api_tags (api_id, tag) VALUES
    ->     ((SELECT api_id FROM apis WHERE name = 'Translation API'),       'nlp'),
    ->     ((SELECT api_id FROM apis WHERE name = 'Translation API'),       'text'),
    ->     ((SELECT api_id FROM apis WHERE name = 'Image Processing API'),  'vision'),
    ->     ((SELECT api_id FROM apis WHERE name = 'Report Generation API'), 'pdf');
Query OK, 4 rows affected (0.00 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql> -- weak entity: api_versions (PK = api_id + version_name)
mysql> INSERT INTO api_versions (api_id, version_name)
    -> SELECT api_id, 'v1' FROM apis;
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> -- weak entity: api_endpoints (PK = api_id + version_name + http_method + path)
mysql> INSERT INTO api_endpoints (api_id, version_name, http_method, path, description)
    -> SELECT a.api_id, 'v1', 'POST', '/translate', 'Translate text'
    -> FROM apis a WHERE a.name = 'Translation API';
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO api_endpoints (api_id, version_name, http_method, path, description)
    -> SELECT a.api_id, 'v1', 'POST', '/process', 'Process an image'
    -> FROM apis a WHERE a.name = 'Image Processing API';
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO api_endpoints (api_id, version_name, http_method, path, description)
    -> SELECT a.api_id, 'v1', 'POST', '/generate', 'Generate a report'
    -> FROM apis a WHERE a.name = 'Report Generation API';
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO pricing_plans (plan_name, monthly_fee, included_requests, rate_limit_per_minute) VALUES
    ->     ('FREE',        0,    1000,    30),
    ->     ('PRO',         999,  50000,   300),
    ->     ('ENTERPRISE',  4999, 1000000, 3000);
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> -- multivalued attribute: plan features
mysql> INSERT INTO plan_features (plan_id, feature) VALUES
    ->     ((SELECT plan_id FROM pricing_plans WHERE plan_name = 'FREE'),       'Community support'),
    ->     ((SELECT plan_id FROM pricing_plans WHERE plan_name = 'PRO'),        'Email support'),
    ->     ((SELECT plan_id FROM pricing_plans WHERE plan_name = 'PRO'),        'Usage dashboard'),
    ->     ((SELECT plan_id FROM pricing_plans WHERE plan_name = 'ENTERPRISE'), 'Dedicated support'),
    ->     ((SELECT plan_id FROM pricing_plans WHERE plan_name = 'ENTERPRISE'), 'SLA 99.99%');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO api_pricing (api_id, plan_id, price_per_request) VALUES
    ->     ((SELECT api_id FROM apis WHERE name = 'Translation API'),
    ->      (SELECT plan_id FROM pricing_plans WHERE plan_name = 'FREE'), 0.50),
    ->     ((SELECT api_id FROM apis WHERE name = 'Translation API'),
    ->      (SELECT plan_id FROM pricing_plans WHERE plan_name = 'PRO'),  0.30),
    ->     ((SELECT api_id FROM apis WHERE name = 'Image Processing API'),
    ->      (SELECT plan_id FROM pricing_plans WHERE plan_name = 'FREE'), 2.00),
    ->     ((SELECT api_id FROM apis WHERE name = 'Image Processing API'),
    ->      (SELECT plan_id FROM pricing_plans WHERE plan_name = 'PRO'),  1.50),
    ->     ((SELECT api_id FROM apis WHERE name = 'Report Generation API'),
    ->      (SELECT plan_id FROM pricing_plans WHERE plan_name = 'FREE'), 5.00),
    ->     ((SELECT api_id FROM apis WHERE name = 'Report Generation API'),
    ->      (SELECT plan_id FROM pricing_plans WHERE plan_name = 'PRO'),  3.50);
Query OK, 6 rows affected (0.00 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO subscriptions (organization_id, plan_id, start_date, status) VALUES
    ->     ((SELECT organization_id FROM organizations WHERE name = 'Acme Technologies'),
    ->      (SELECT plan_id FROM pricing_plans WHERE plan_name = 'PRO'),
    ->      CURDATE(), 'ACTIVE');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> INSERT INTO api_access (organization_id, api_id)
    -> SELECT o.organization_id, a.api_id
    -> FROM organizations o
    -> JOIN apis a ON a.organization_id = o.organization_id
    -> WHERE o.name = 'Acme Technologies';
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> -- the audit trigger writes a CREATED row automatically
mysql> INSERT INTO api_keys (organization_id, key_prefix, key_hash, expires_at) VALUES
    ->     ((SELECT organization_id FROM organizations WHERE name = 'Acme Technologies'),
    ->      'ak_live_abc', 'example_secure_hash_value',
    ->      CURRENT_TIMESTAMP + INTERVAL 1 YEAR);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> -- multivalued attribute: key scopes
mysql> INSERT INTO api_key_scopes (api_key_id, scope) VALUES
    ->     ((SELECT api_key_id FROM api_keys WHERE key_prefix = 'ak_live_abc'), 'read'),
    ->     ((SELECT api_key_id FROM api_keys WHERE key_prefix = 'ak_live_abc'), 'write');
Query OK, 2 rows affected (0.00 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO usage_records
    ->     (organization_id, api_id, subscription_id, usage_date,
    ->      request_count, billable_requests, amount)
    -> VALUES
    ->     ((SELECT organization_id FROM organizations WHERE name = 'Acme Technologies'),
    ->      (SELECT api_id FROM apis WHERE name = 'Translation API'),
    ->      (SELECT subscription_id FROM subscriptions
    ->       WHERE status = 'ACTIVE'
    ->         AND organization_id = (SELECT organization_id FROM organizations
    ->                                WHERE name = 'Acme Technologies')
    ->       LIMIT 1),
    ->      CURDATE(), 2000, 1000, 300);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> /*
   /*> ========================================================================
   /*> 27. INVOICE + INVOICE ITEMS (transaction, weak-entity numbering, triggers)
   /*> ========================================================================
   /*> */
mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO invoices
    ->     (organization_id, subscription_id, billing_period_start,
    ->      billing_period_end, tax, status, due_date)
    -> VALUES
    ->     ((SELECT organization_id FROM organizations WHERE name = 'Acme Technologies'),
    ->      (SELECT subscription_id FROM subscriptions
    ->       WHERE status = 'ACTIVE'
    ->         AND organization_id = (SELECT organization_id FROM organizations
    ->                                WHERE name = 'Acme Technologies')
    ->       LIMIT 1),
    ->      DATE_FORMAT(CURDATE(), '%Y-%m-01'),
    ->      LAST_DAY(CURDATE()),
    ->      0,
    ->      'DRAFT',
    ->      CURDATE() + INTERVAL 30 DAY);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> -- MySQL does not allow a statement to read a table (invoices) that a trigger
mysql> -- fired by the same statement updates, so the invoice id goes into a variable.
mysql> SET @new_invoice_id = (SELECT MAX(invoice_id) FROM invoices);
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> -- invoice_item_id is omitted (defaults to 0) -> trigger assigns 1, then 2
mysql> INSERT INTO invoice_items (invoice_id, api_id, request_count, unit_price, amount)
    -> VALUES
    ->     (@new_invoice_id,
    ->      (SELECT api_id FROM apis WHERE name = 'Translation API'),
    ->      1000, 0.30, 300.00);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> INSERT INTO invoice_items (invoice_id, api_id, request_count, unit_price, amount)
    -> VALUES
    ->     (@new_invoice_id,
    ->      (SELECT api_id FROM apis WHERE name = 'Image Processing API'),
    ->      500, 1.50, 750.00);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> /*
   /*> ========================================================================
   /*> 29. UPDATE / DELETE DEMONSTRATIONS
   /*> ========================================================================
   /*> */
mysql>
mysql> -- Revoke an API key -> trigger writes a REVOKED audit row
mysql> UPDATE api_keys SET revoked_at = CURRENT_TIMESTAMP WHERE key_prefix = 'ak_live_abc';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT audit_id, api_key_id, action, action_time, details
    -> FROM api_key_audit
    -> ORDER BY audit_id DESC;
+----------+------------+---------+---------------------+-----------------+
| audit_id | api_key_id | action  | action_time         | details         |
+----------+------------+---------+---------------------+-----------------+
|        2 |          1 | REVOKED | 2026-10-03 20:15:54 | API key revoked |
|        1 |          1 | CREATED | 2026-10-03 20:15:54 | API key created |
+----------+------------+---------+---------------------+-----------------+
2 rows in set (0.00 sec)

mysql>
mysql> -- INSERT / UPDATE / DELETE inside a transaction that is rolled back
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO organizations (name) VALUES ('DML Demonstration Organization');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> UPDATE organizations
    -> SET name = 'Updated DML Demonstration Organization'
    -> WHERE name = 'DML Demonstration Organization';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> DELETE FROM organizations
    -> WHERE name = 'Updated DML Demonstration Organization';
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> -- Weak-entity cascade demonstration (rolled back): deleting an API removes
mysql> -- its versions and endpoints automatically. Rows 2-3 show the cascade.
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT COUNT(*) AS versions_before  FROM api_versions;
+-----------------+
| versions_before |
+-----------------+
|               3 |
+-----------------+
1 row in set (0.00 sec)
```

```
mysql> SELECT COUNT(*) AS endpoints_before FROM api_endpoints;
+------------------+
| endpoints_before |
+------------------+
|                3 |
+------------------+
1 row in set (0.00 sec)

mysql>
mysql> DELETE FROM api_pricing;
Query OK, 6 rows affected (0.00 sec)

mysql> DELETE FROM api_access;
Query OK, 3 rows affected (0.00 sec)

mysql> DELETE FROM api_tags;
Query OK, 4 rows affected (0.00 sec)

mysql> DELETE FROM apis WHERE name = 'Report Generation API';
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT COUNT(*) AS versions_after   FROM api_versions;
+----------------+
| versions_after |
+----------------+
|              2 |
+----------------+
1 row in set (0.00 sec)

mysql> SELECT COUNT(*) AS endpoints_after  FROM api_endpoints;
+-----------------+
| endpoints_after |
+-----------------+
|               2 |
+-----------------+
1 row in set (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)
```

SQL QUERIES

```
mysql> /*
   /*> ========================================================================
   /*> 28. QUERIES
   /*> ========================================================================
   /*> */
mysql>
mysql> -- 28.1 Basic selects
mysql> SELECT * FROM roles;
+---------+-----------+-------------------------+
| role_id | role_name | description             |
+---------+-----------+-------------------------+
|       1 | ADMIN     | Platform administrator  |
|       2 | DEVELOPER | API consumer/developer  |
+---------+-----------+-------------------------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM users;
+---------+----------------------+------------------+-------------------------------+---------+-----------+---------------------+---------------------+
| user_id | name                 | email            | password_hash                 | role_id | is_active | created_at          | updated_at          |
+---------+----------------------+------------------+-------------------------------+---------+-----------+---------------------+---------------------+
|       1 | System Administrator | admin@example.com | $2b$12$example_admin_hash     |       1 |         1 | 2026-10-03 20:15:54 | 2026-10-03 20:15:54 |
|       2 | John Developer       | john@example.com  | $2b$12$example_developer_hash |       2 |         1 | 2026-10-03 20:15:54 | 2026-10-03 20:15:54 |
+---------+----------------------+------------------+-------------------------------+---------+-----------+---------------------+---------------------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM organizations;
+-----------------+-------------------+---------------------+
| organization_id | name              | created_at          |
+-----------------+-------------------+---------------------+
|               1 | Acme Technologies | 2026-10-03 20:15:54 |
|               2 | DevCloud Solutions | 2026-10-03 20:15:54 |
+-----------------+-------------------+---------------------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM apis;
+--------+-----------------+----------------------+---------------------+-----------------------------------+--------+---------------------+---------------------+
| api_id | organization_id | name                 | description         | base_url                          | status | created_at          | updated_at          |
+--------+-----------------+----------------------+---------------------+-----------------------------------+--------+---------------------+---------------------+
|      1 |               1 | Translation API      | Text translation service | https://api.example.com/translate | ACTIVE | 2026-10-03 20:15:54 | 2026-10-03 20:15:54 |
|      2 |               1 | Image Processing API | Image processing service | https://api.example.com/image     | ACTIVE | 2026-10-03 20:15:54 | 2026-10-03 20:15:54 |
|      3 |               1 | Report Generation API | Generates PDF reports | https://api.example.com/reports   | ACTIVE | 2026-10-03 20:15:54 | 2026-10-03 20:15:54 |
+--------+-----------------+----------------------+---------------------+-----------------------------------+--------+---------------------+---------------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM pricing_plans;
+---------+-----------+-------------+--------------------+---------------------+---------------------+
| plan_id | plan_name | monthly_fee | included_requests  | rate_limit_per_minute | created_at          |
+---------+-----------+-------------+--------------------+---------------------+---------------------+
|       1 | FREE      |        0.00 |               1000 |                  30 | 2026-10-03 20:15:54 |
|       2 | PRO       |      999.00 |              50000 |                 300 | 2026-10-03 20:15:54 |
|       3 | ENTERPRISE |     4999.00 |            1000000 |                3000 | 2026-10-03 20:15:54 |
+---------+-----------+-------------+--------------------+---------------------+---------------------+
3 rows in set (0.00 sec)
```

```
mysql> -- 28.2 Weak entity join: API -> version -> endpoint (composite key join)
mysql> SELECT a.name AS api_name,
    ->        av.version_name,
    ->        ae.http_method,
    ->        ae.path,
    ->        ae.description
    -> FROM api_endpoints ae
    -> JOIN api_versions av
    ->      ON  ae.api_id       = av.api_id
    ->      AND ae.version_name = av.version_name
    -> JOIN apis a
    ->      ON av.api_id = a.api_id
    -> ORDER BY a.name;
+----------------------+--------------+-------------+------------+--------------------+
| api_name             | version_name | http_method | path       | description        |
+----------------------+--------------+-------------+------------+--------------------+
| Image Processing API | v1           | POST        | /process   | Process an image   |
| Report Generation API | v1           | POST        | /generate  | Generate a report  |
| Translation API      | v1           | POST        | /translate | Translate text     |
+----------------------+--------------+-------------+------------+--------------------+
3 rows in set (0.00 sec)

mysql>
mysql> -- 28.3 Weak entity join: invoice -> numbered items
mysql> SELECT i.invoice_id,
    ->        ii.invoice_item_id AS line_no,
    ->        a.name             AS api_name,
    ->        ii.request_count,
    ->        ii.unit_price,
    ->        ii.amount
    -> FROM invoices i
    -> JOIN invoice_items ii ON ii.invoice_id = i.invoice_id
    -> JOIN apis a           ON a.api_id = ii.api_id
    -> ORDER BY i.invoice_id, ii.invoice_item_id;
+------------+---------+----------------------+---------------+------------+--------+
| invoice_id | line_no | api_name             | request_count | unit_price | amount |
+------------+---------+----------------------+---------------+------------+--------+
|          1 |       1 | Translation API      |          1000 |     0.3000 | 300.00 |
|          1 |       2 | Image Processing API |           500 |     1.5000 | 750.00 |
+------------+---------+----------------------+---------------+------------+--------+
2 rows in set (0.00 sec)

mysql>
mysql> -- 28.4 Trigger-calculated invoice totals
mysql> SELECT invoice_id, subtotal, tax, total_amount, status
    -> FROM invoices
    -> ORDER BY invoice_id DESC;
+------------+----------+------+--------------+--------+
| invoice_id | subtotal | tax  | total_amount | status |
+------------+----------+------+--------------+--------+
|          1 |  1050.00 | 0.00 |      1050.00 | DRAFT  |
+------------+----------+------+--------------+--------+
1 row in set (0.00 sec)
```

```
mysql> -- 28.5 Multivalued attributes
mysql> SELECT u.name, GROUP_CONCAT(p.phone_number ORDER BY p.phone_number SEPARATOR ', ') AS phone_numbers
    -> FROM users u
    -> LEFT JOIN user_phone_numbers p ON p.user_id = u.user_id
    -> GROUP BY u.user_id, u.name;
+----------------------+----------------------------------+
| name                 | phone_numbers                    |
+----------------------+----------------------------------+
| System Administrator | +91-9000000001, +91-9000000002 |
| John Developer       | +91-9000000003                   |
+----------------------+----------------------------------+
2 rows in set (0.00 sec)

mysql>
mysql> SELECT a.name AS api_name, GROUP_CONCAT(t.tag ORDER BY t.tag SEPARATOR ', ') AS tags
    -> FROM apis a
    -> LEFT JOIN api_tags t ON t.api_id = a.api_id
    -> GROUP BY a.api_id, a.name;
+----------------------+-----------+
| api_name             | tags      |
+----------------------+-----------+
| Translation API      | nlp, text |
| Image Processing API | vision    |
| Report Generation API | pdf       |
+----------------------+-----------+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT pp.plan_name, GROUP_CONCAT(pf.feature ORDER BY pf.feature SEPARATOR ', ') AS features
    -> FROM pricing_plans pp
    -> LEFT JOIN plan_features pf ON pf.plan_id = pp.plan_id
    -> GROUP BY pp.plan_id, pp.plan_name;
+------------+---------------------------------+
| plan_name  | features                        |
+------------+---------------------------------+
| FREE       | Community support               |
| PRO        | Email support, Usage dashboard |
| ENTERPRISE | Dedicated support, SLA 99.99%  |
+------------+---------------------------------+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT k.key_prefix, GROUP_CONCAT(s.scope ORDER BY s.scope SEPARATOR ', ') AS scopes
    -> FROM api_keys k
    -> LEFT JOIN api_key_scopes s ON s.api_key_id = k.api_key_id
    -> GROUP BY k.api_key_id, k.key_prefix;
+-------------+-------------+
| key_prefix  | scopes      |
+-------------+-------------+
| ak_live_abc | read, write |
+-------------+-------------+
1 row in set (0.00 sec)
```

```
mysql> -- 28.6 Organizations, subscriptions and plans
mysql> SELECT o.organization_id, o.name AS organization_name,
    ->        p.plan_name, s.start_date, s.end_date, s.status
    -> FROM organizations o
    -> JOIN subscriptions s  ON o.organization_id = s.organization_id
    -> JOIN pricing_plans p  ON s.plan_id = p.plan_id;
+-----------------+-------------------+-----------+------------+----------+--------+
| organization_id | organization_name | plan_name | start_date | end_date | status |
+-----------------+-------------------+-----------+------------+----------+--------+
|               1 | Acme Technologies | PRO       | 2026-10-03 | NULL     | ACTIVE |
+-----------------+-------------------+-----------+------------+----------+--------+
1 row in set (0.00 sec)

mysql>
mysql> -- 28.7 Organization -> API access (M:N)
mysql> SELECT o.name AS organization_name, a.name AS api_name, aa.status AS access_status
    -> FROM api_access aa
    -> JOIN organizations o ON aa.organization_id = o.organization_id
    -> JOIN apis a          ON aa.api_id = a.api_id
    -> ORDER BY o.name, a.name;
+-------------------+----------------------+---------------+
| organization_name | api_name             | access_status |
+-------------------+----------------------+---------------+
| Acme Technologies | Image Processing API  | ACTIVE        |
| Acme Technologies | Report Generation API | ACTIVE        |
| Acme Technologies | Translation API       | ACTIVE        |
+-------------------+----------------------+---------------+
3 rows in set (0.00 sec)

mysql>
mysql> -- 28.8 GROUP BY / HAVING / aggregates
mysql> SELECT a.name AS api_name, SUM(u.request_count) AS total_requests
    -> FROM usage_records u
    -> JOIN apis a ON u.api_id = a.api_id
    -> GROUP BY a.api_id, a.name
    -> ORDER BY total_requests DESC;
+-----------------+----------------+
| api_name        | total_requests |
+-----------------+----------------+
| Translation API |           2000 |
+-----------------+----------------+
1 row in set (0.00 sec)

mysql>
mysql> SELECT a.name AS api_name, SUM(u.request_count) AS total_requests
    -> FROM usage_records u
    -> JOIN apis a ON u.api_id = a.api_id
    -> GROUP BY a.api_id, a.name
    -> HAVING SUM(u.request_count) > 1000;
+-----------------+----------------+
| api_name        | total_requests |
+-----------------+----------------+
| Translation API |           2000 |
+-----------------+----------------+
1 row in set (0.00 sec)
```

```
mysql>
mysql> SELECT COUNT(*)                AS total_usage_records,
    ->        SUM(request_count)      AS total_requests,
    ->        SUM(billable_requests)  AS total_billable_requests,
    ->        SUM(amount)             AS total_usage_amount,
    ->        AVG(amount)             AS average_usage_amount,
    ->        MIN(amount)             AS minimum_usage_amount,
    ->        MAX(amount)             AS maximum_usage_amount
    -> FROM usage_records;
+---------------------+----------------+--------------------------+---------------------+---------------------+---------------------+---------------------+
| total_usage_records | total_requests | total_billable_requests | total_usage_amount | average_usage_amount | minimum_usage_amount | maximum_usage_amount |
+---------------------+----------------+--------------------------+---------------------+---------------------+---------------------+---------------------+
|                   1 |           2000 |                     1000 |              300.00 |          300.000000 |              300.00 |              300.00 |
+---------------------+----------------+--------------------------+---------------------+---------------------+---------------------+---------------------+
1 row in set (0.00 sec)

mysql>
mysql> -- 28.9 Subqueries
mysql> SELECT name
    -> FROM organizations
    -> WHERE organization_id IN (
    ->     SELECT organization_id
    ->     FROM subscriptions
    ->     WHERE plan_id = (SELECT plan_id FROM pricing_plans WHERE plan_name = 'PRO')
    -> );
+------------------+
| name             |
+------------------+
| Acme Technologies |
+------------------+
1 row in set (0.00 sec)

mysql>
mysql> SELECT a.name AS api_name, ap.price_per_request
    -> FROM api_pricing ap
    -> JOIN apis a ON ap.api_id = a.api_id
    -> WHERE ap.price_per_request > (SELECT AVG(price_per_request) FROM api_pricing);
+----------------------+-------------------+
| api_name             | price_per_request |
+----------------------+-------------------+
| Report Generation API |            5.0000 |
| Report Generation API |            3.5000 |
+----------------------+-------------------+
2 rows in set (0.00 sec)

mysql>
mysql> -- 28.10 ORDER BY + LIMIT
mysql> SELECT a.name AS api_name, ap.price_per_request
    -> FROM api_pricing ap
    -> JOIN apis a ON ap.api_id = a.api_id
    -> ORDER BY ap.price_per_request DESC
    -> LIMIT 5;
+----------------------+-------------------+
| api_name             | price_per_request |
+----------------------+-------------------+
| Report Generation API |            5.0000 |
| Report Generation API |            3.5000 |
| Image Processing API  |            2.0000 |
| Image Processing API  |            1.5000 |
| Translation API       |            0.5000 |
+----------------------+-------------------+
5 rows in set (0.00 sec)
```

```
mysql> -- 28.11 API key status
mysql> SELECT o.name AS organization_name, ak.key_prefix, ak.created_at,
    ->        ak.expires_at, ak.revoked_at,
    ->        CASE
    ->            WHEN ak.revoked_at IS NOT NULL THEN 'REVOKED'
    ->            WHEN ak.expires_at IS NOT NULL
    ->                 AND ak.expires_at < CURRENT_TIMESTAMP THEN 'EXPIRED'
    ->            ELSE 'ACTIVE'
    ->        END AS key_status
    -> FROM api_keys ak
    -> JOIN organizations o ON ak.organization_id = o.organization_id;
+-------------------+-------------+---------------------+---------------------+------------+------------+
| organization_name | key_prefix  | created_at          | expires_at          | revoked_at | key_status |
+-------------------+-------------+---------------------+---------------------+------------+------------+
| Acme Technologies | ak_live_abc | 2026-10-03 20:15:54 | 2027-10-03 20:15:54 | NULL       | ACTIVE     |
+-------------------+-------------+---------------------+---------------------+------------+------------+
1 row in set (0.00 sec)

mysql>
mysql> -- 28.12 Revenue and billing
mysql> SELECT a.name AS api_name, SUM(ii.amount) AS revenue
    -> FROM invoice_items ii
    -> JOIN apis a ON ii.api_id = a.api_id
    -> GROUP BY a.api_id, a.name
    -> ORDER BY revenue DESC;
+----------------------+---------+
| api_name             | revenue |
+----------------------+---------+
| Image Processing API |  750.00 |
| Translation API      |  300.00 |
+----------------------+---------+
2 rows in set (0.00 sec)

mysql>
mysql> SELECT DATE_FORMAT(generated_at, '%Y-%m-01') AS month,
    ->        SUM(total_amount) AS revenue
    -> FROM invoices
    -> WHERE status IN ('ISSUED', 'PAID', 'OVERDUE')
    -> GROUP BY DATE_FORMAT(generated_at, '%Y-%m-01')
    -> ORDER BY month DESC;
Empty set (0.00 sec)

mysql>
mysql> -- 28.13 NOT EXISTS
mysql> SELECT o.organization_id, o.name
    -> FROM organizations o
    -> WHERE NOT EXISTS (
    ->     SELECT 1 FROM subscriptions s
    ->     WHERE s.organization_id = o.organization_id AND s.status = 'ACTIVE'
    -> );
+-----------------+---------------------+
| organization_id | name                |
+-----------------+---------------------+
|               2 | DevCloud Solutions |
+-----------------+---------------------+
1 row in set (0.00 sec)
```

```
mysql>
mysql> SELECT a.api_id, a.name
    -> FROM apis a
    -> WHERE NOT EXISTS (
    ->     SELECT 1 FROM api_access aa
    ->     WHERE aa.api_id = a.api_id AND aa.status = 'ACTIVE'
    -> );
Empty set (0.00 sec)

mysql>
mysql> -- 28.14 Views
mysql> SELECT * FROM api_usage_summary ORDER BY total_requests DESC;
+--------+-----------------+----------------+-------------------------+--------------+---------------------+
| api_id | api_name        | total_requests | total_billable_requests | total_amount | average_usage_amount |
+--------+-----------------+----------------+-------------------------+--------------+---------------------+
|      1 | Translation API |           2000 |                    1000 |       300.00 |          300.000000 |
+--------+-----------------+----------------+-------------------------+--------------+---------------------+
1 row in set (0.00 sec)

mysql> SELECT * FROM monthly_api_usage ORDER BY month DESC;
+-----------------+-------------------+--------+-----------------+------------+----------------+------------------+--------------+
| organization_id | organization_name | api_id | api_name        | month      | total_requests | billable_requests | total_amount |
+-----------------+-------------------+--------+-----------------+------------+----------------+------------------+--------------+
|               1 | Acme Technologies |      1 | Translation API | 2026-10-01 |           2000 |             1000 |       300.00 |
+-----------------+-------------------+--------+-----------------+------------+----------------+------------------+--------------+
1 row in set (0.00 sec)

mysql> SELECT * FROM organization_billing_summary ORDER BY total_billed DESC;
+-----------------+--------------------+---------------+----------+------+---------------+
| organization_id | organization_name  | invoice_count | subtotal | tax  | total_billed |
+-----------------+--------------------+---------------+----------+------+---------------+
|               1 | Acme Technologies  |             1 |  1050.00 | 0.00 |       1050.00 |
|               2 | DevCloud Solutions |             0 |     0.00 | 0.00 |          0.00 |
+-----------------+--------------------+---------------+----------+------+---------------+
2 rows in set (0.00 sec)

mysql>
mysql> -- 28.15 Functions
mysql> SELECT calculate_usage_cost(
    ->            (SELECT organization_id FROM organizations WHERE name = 'Acme Technologies'),
    ->            (SELECT api_id FROM apis WHERE name = 'Translation API'),
    ->            5000) AS calculated_cost;
+-----------------+
| calculated_cost |
+-----------------+
|            0.00 |
+-----------------+
1 row in set (0.00 sec)

mysql>
mysql> SELECT calculate_billable_requests(
    ->            (SELECT organization_id FROM organizations WHERE name = 'Acme Technologies'),
    ->            5000) AS billable_requests;
+-------------------+
| billable_requests |
+-------------------+
|                 0 |
+-------------------+
1 row in set (0.00 sec)
```

```
mysql>
mysql> /*
   /*> ============================================================================
   /*> 30. FINAL DATABASE CHECK
   /*> ============================================================================
   /*> */
mysql>
mysql> SELECT 'roles' AS table_name, COUNT(*) AS row_count FROM roles
    -> UNION ALL SELECT 'users',              COUNT(*) FROM users
    -> UNION ALL SELECT 'user_phone_numbers', COUNT(*) FROM user_phone_numbers
    -> UNION ALL SELECT 'organizations',      COUNT(*) FROM organizations
    -> UNION ALL SELECT 'organization_members', COUNT(*) FROM organization_members
    -> UNION ALL SELECT 'apis',               COUNT(*) FROM apis
    -> UNION ALL SELECT 'api_tags',           COUNT(*) FROM api_tags
    -> UNION ALL SELECT 'api_versions',       COUNT(*) FROM api_versions
    -> UNION ALL SELECT 'api_endpoints',      COUNT(*) FROM api_endpoints
    -> UNION ALL SELECT 'pricing_plans',      COUNT(*) FROM pricing_plans
    -> UNION ALL SELECT 'plan_features',      COUNT(*) FROM plan_features
    -> UNION ALL SELECT 'api_pricing',        COUNT(*) FROM api_pricing
    -> UNION ALL SELECT 'subscriptions',      COUNT(*) FROM subscriptions
    -> UNION ALL SELECT 'api_keys',           COUNT(*) FROM api_keys
    -> UNION ALL SELECT 'api_key_scopes',     COUNT(*) FROM api_key_scopes
    -> UNION ALL SELECT 'api_access',         COUNT(*) FROM api_access
    -> UNION ALL SELECT 'usage_records',      COUNT(*) FROM usage_records
    -> UNION ALL SELECT 'invoices',           COUNT(*) FROM invoices
    -> UNION ALL SELECT 'invoice_items',      COUNT(*) FROM invoice_items
    -> UNION ALL SELECT 'api_key_audit',      COUNT(*) FROM api_key_audit;
+---------------------+-----------+
| table_name          | row_count |
+---------------------+-----------+
| roles               |         2 |
| users               |         2 |
| user_phone_numbers  |         3 |
| organizations       |         2 |
| organization_members |         2 |
| apis                |         3 |
| api_tags            |         4 |
| api_versions        |         3 |
| api_endpoints       |         3 |
| pricing_plans       |         3 |
| plan_features       |         5 |
| api_pricing         |         6 |
| subscriptions       |         1 |
| api_keys            |         1 |
| api_key_scopes      |         2 |
| api_access          |         3 |
| usage_records       |         1 |
| invoices            |         1 |
| invoice_items       |         2 |
| api_key_audit       |         2 |
+---------------------+-----------+
20 rows in set (0.01 sec)
```